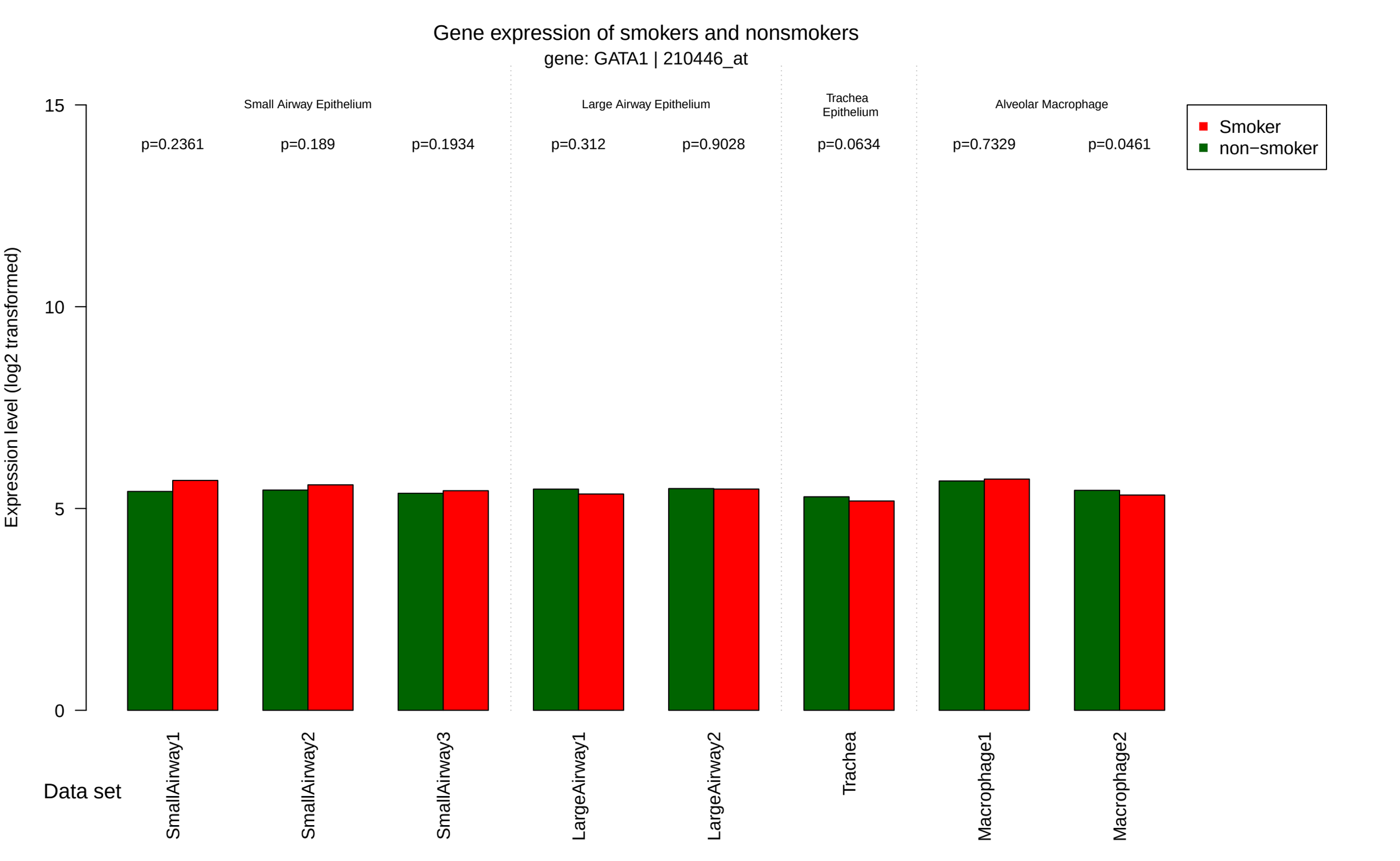
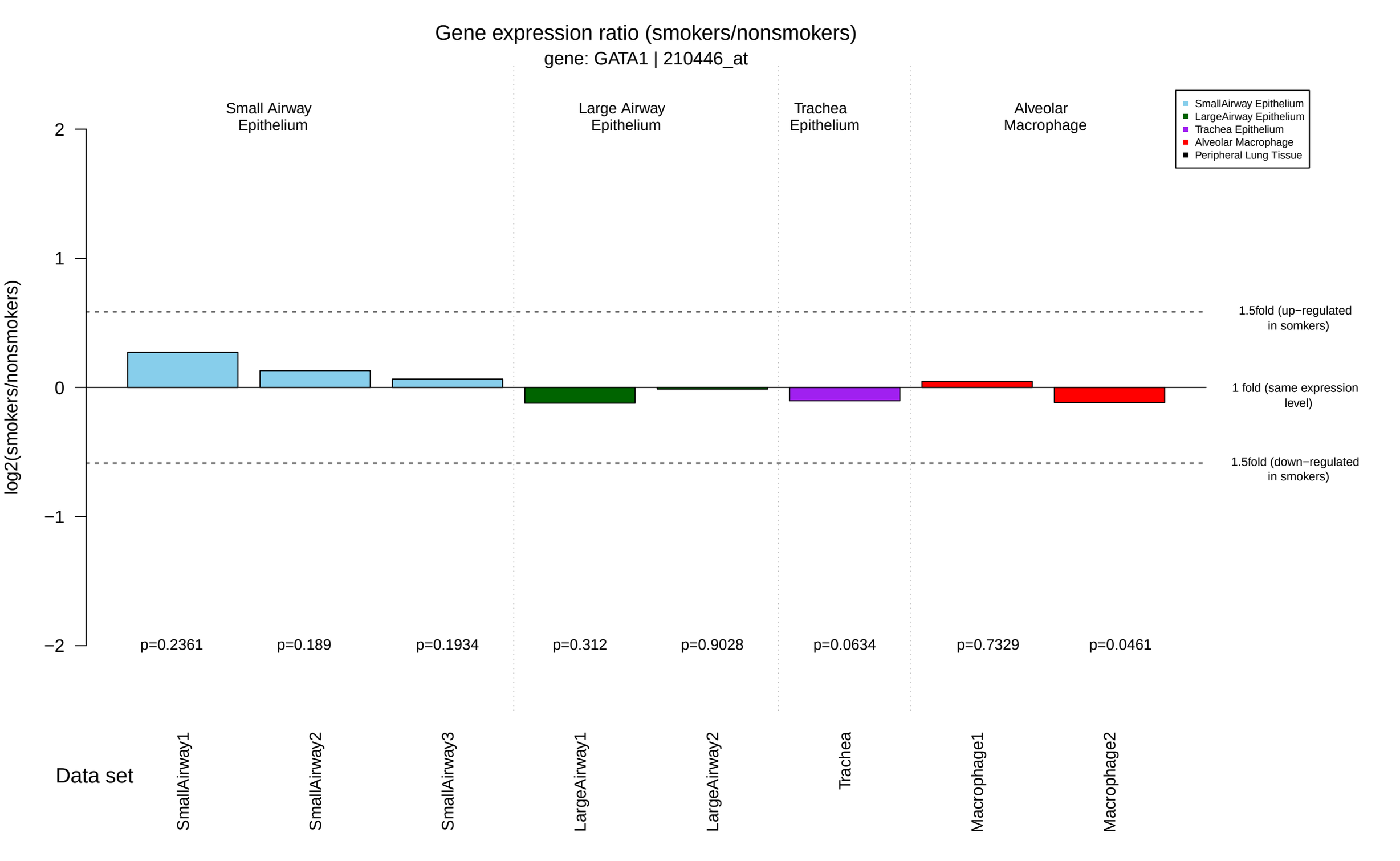
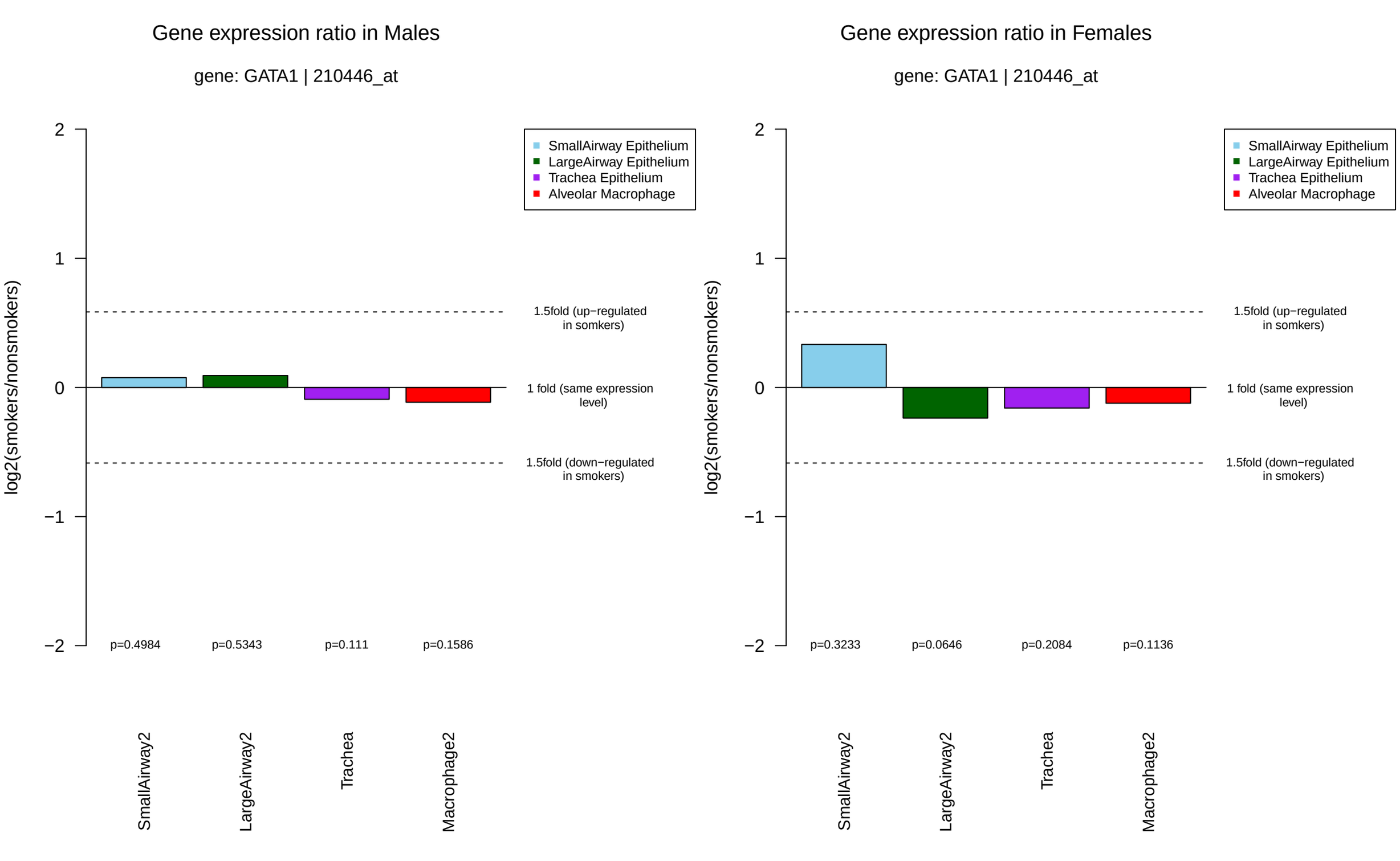
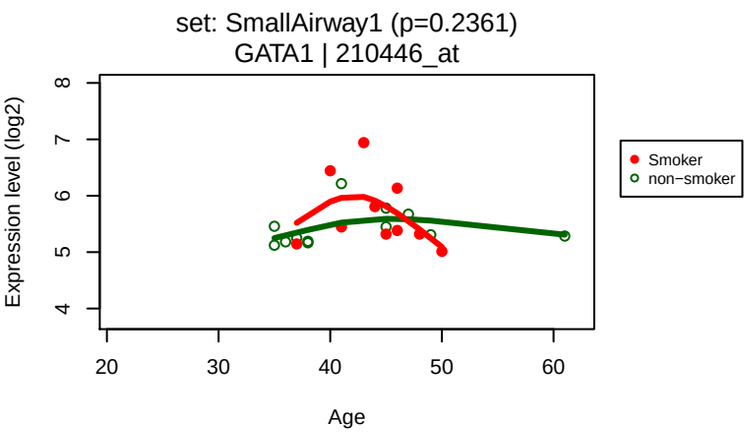
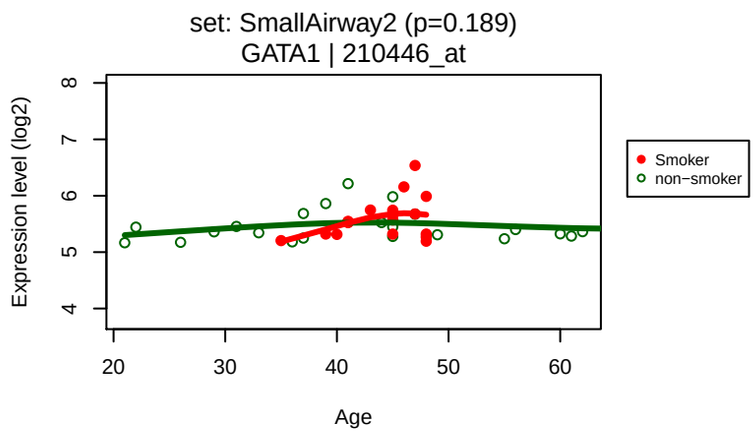
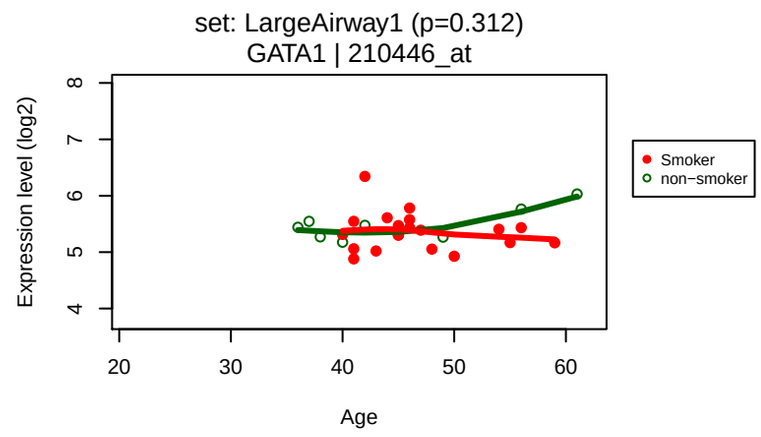

Gene expression of smokers and nonsmokers
gene: GATA1 | 210446_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2361
p=0.189
p=0.1934
p=0.312
p=0.9028
p=0.0634
p=0.7329
p=0.0461
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

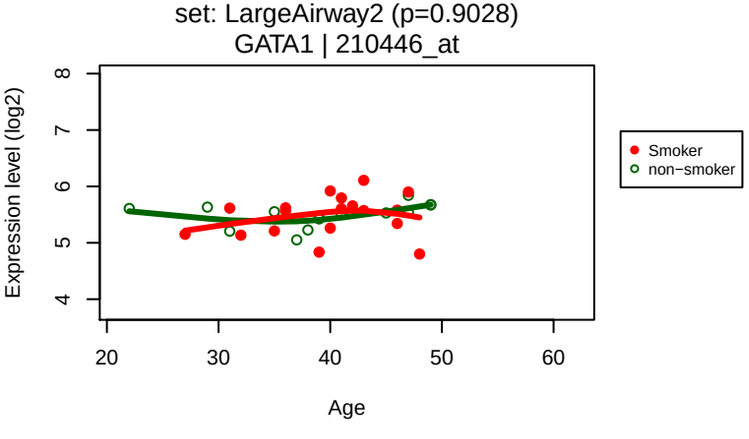

Gene expression ratio (smokers/nonsmokers)
gene: GATA1 | 210446_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2361
p=0.189
p=0.1934
p=0.312
p=0.9028
p=0.0634
p=0.7329
p=0.0461
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

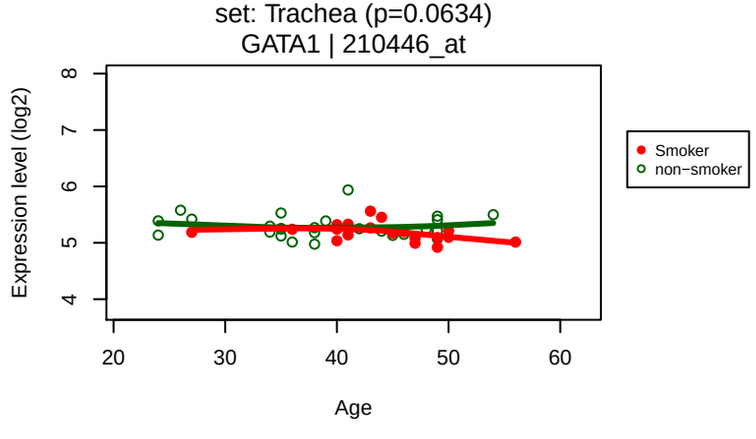

Gene expression ratio in Males
gene: GATA1 | 210446_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4984
p=0.5343
p=0.111
p=0.1586
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATA1 | 210446_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3233
p=0.0646
p=0.2084
p=0.1136
SmallAirway2
LargeAirway2
Trachea
Macrophage2

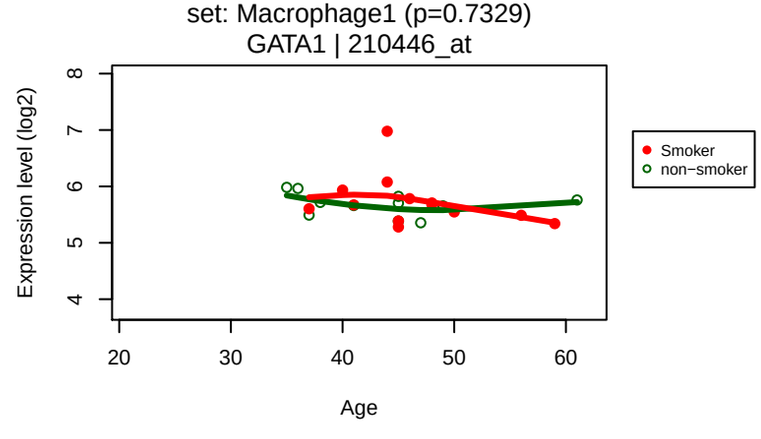

set: SmallAirway1 (p=0.2361)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker

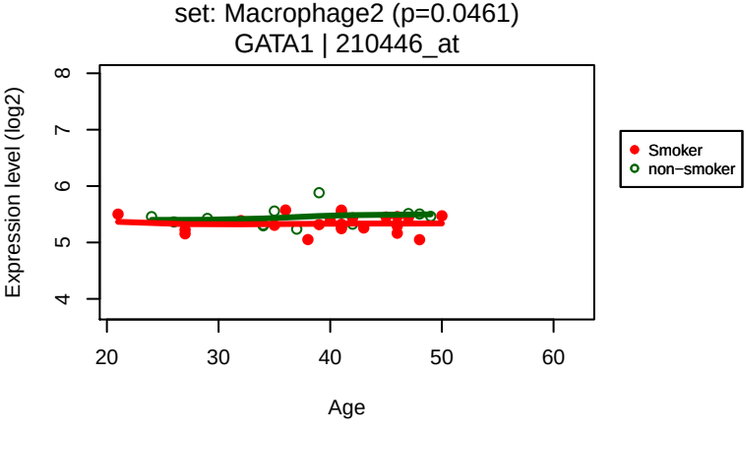

set: SmallAirway2 (p=0.189)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker

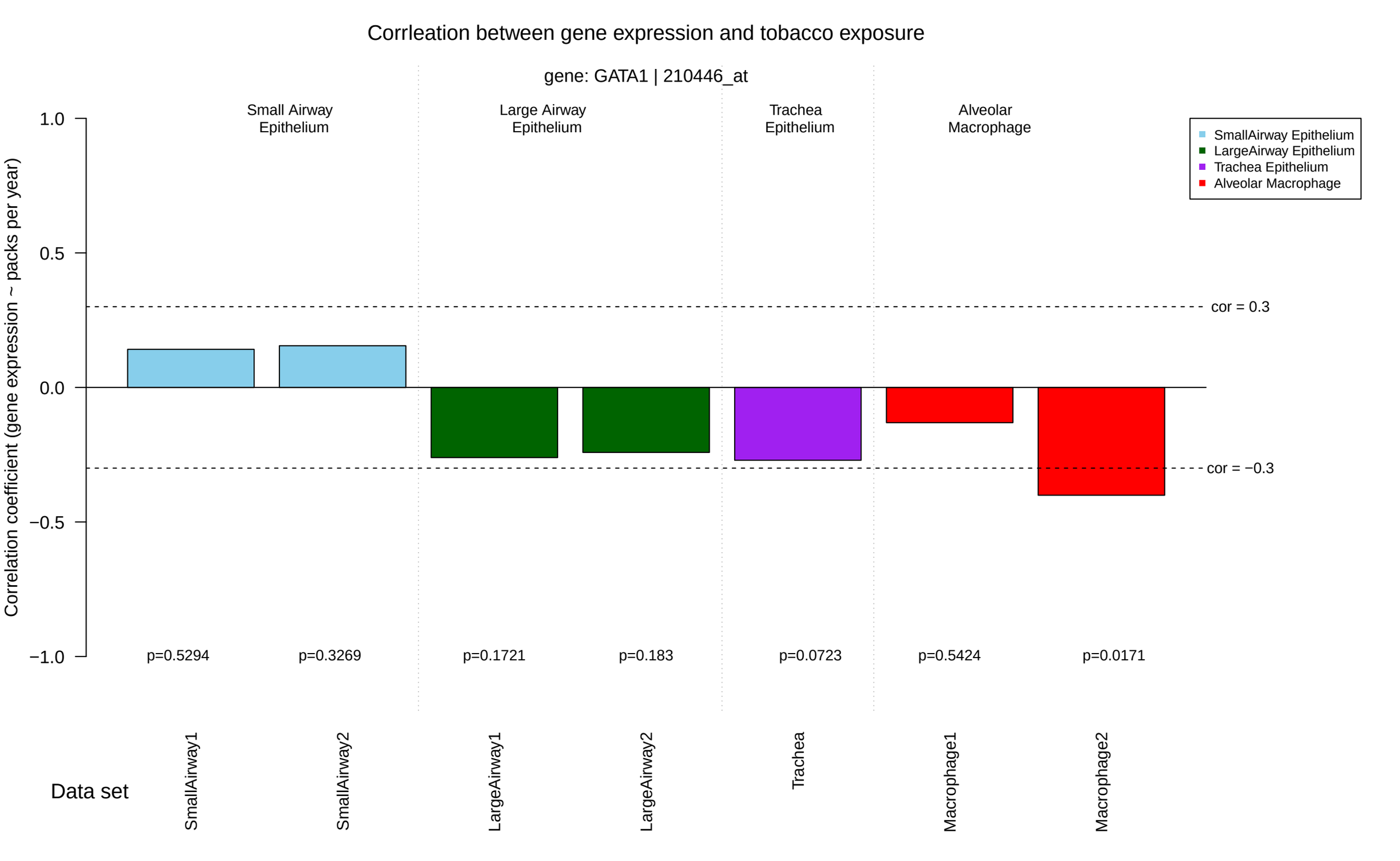

set: LargeAirway1 (p=0.312)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9028)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0634)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7329)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0461)
GATA1 | 210446_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATA1 | 210446_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5294
p=0.3269
p=0.1721
p=0.183
p=0.0723
p=0.5424
p=0.0171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2